higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 3

(6021033)

21 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL sketches must be neat and in proportion.
5. ALL the work that you do NOT want to be marked must be clearly crossed out.
6. Start each question on a NEW page.
7. Write neatly and legibly.

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QUESTION 1

1.1 Choose a description from COLUMN B that matches an item in COLUMN A. Write only the letter (A–K) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1.1	Collapsible spacer	A	activates the starter motor
1.1.2	Rotor arm	B	part of a rear suspension system
1.1.3	Stator	C	part of the gearbox selector mechanism
1.1.4	Solenoid	D	should not be re-used in the differential
1.1.5	Reservoir		
1.1.6	Airflow sensor	E	a container for the brake fluid
1.1.7	Shift fork	F	check if a cylinder is out of round
1.1.8	Ovality	G	a type of tyre construction
1.1.9	Cupping spots	H	a part of the alternator
1.1.10	Leaf spring	I	tyre wear pattern
		J	the link between the coil HT cable and plug cables
		K	measures the intake air

(10 × 1) (10)

1.2 Explain why the cylinders or combustion chambers of diesel engines must be pre-heated on cold start. (3)

1.3 List THREE tests that should be performed on diesel injectors. (3)

1.4 Briefly explain the term *bleeding the brakes.* (1)

1.5 Give TWO possible causes for each of the following brake conditions:

1.5.1 Brakes shuddering.

1.5.2 Brakes dragging.

(2 × 2) (4)

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1.6 Explain FOUR steps that one should follow to remove a wheel cylinder of a drum brake system after the wheels have been removed. (4)

1.7 List THREE functions of the synchroniser unit of a manual transmission. (3)

1.8 Use the picture of the section of a crown wheel in FIGURE 1 below to answer the question.

Identify the tooth contact patterns that are labelled A, B and C.

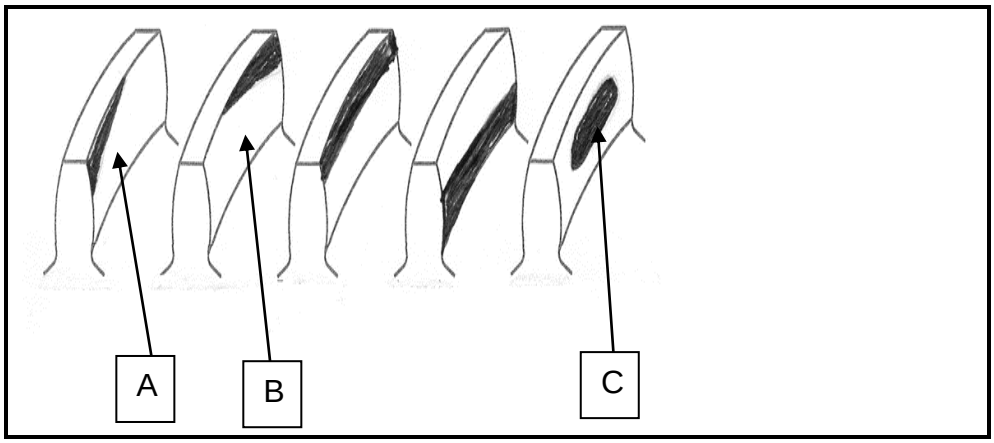


FIGURE 1

(3 × 1) (3)
[31]

QUESTION 2

2.1 Incorrect wheel alignment has a negative effect on the vehicle. Make a simple sketch of a wheel and strut assembly that indicates negative camber. (3)

2.2 Explain THREE reasons for the upper and lower control arms to be unequal in length. (3)

2.3 Briefly explain TWO functions of each of the following suspension components.

2.3.1 The shock absorbers

2.3.2 The coil springs

(2 × 2) (4)

2.4 Name ONE visual inspection that would indicate that the wheel alignment needs attention. (1)

2.5 Name TWO types of balancing weights used to balance different types of wheels. (2)

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2.6 List THREE inspections that one should do on the rim or tyre before it can be balanced. (3)

2.7 TWO types of balancing affect the smooth running of the wheel.

Name ONE that rectifies the sideways wobble of the wheel. (1)

2.8 Describe FOUR effects that will result from an unbalanced wheel fitted to the front of a vehicle. (4)

[21]

QUESTION 3

3.1 Refer to FIGURE 2 below to answer the questions.

3.1.1 Name TWO parts where electro-magnetism is used. (2)

3.1.2 Name the parts ✵A–C.

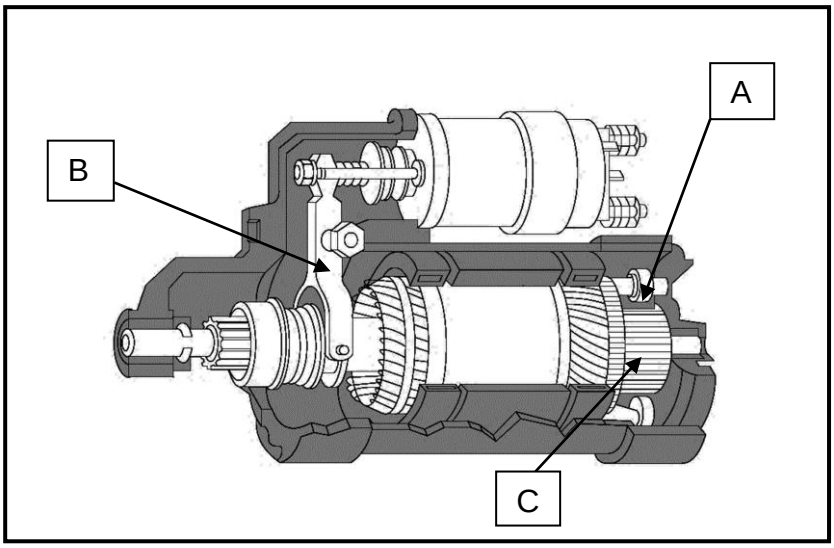


FIGURE 2

(3 × 1) (3)

3.2 Give ONE possible reason for each of the following starter conditions:

3.2.1 The starter swings but does not crank the engine. (1)

3.2.2 The starter solenoid engages the pinion but it does not swing. (1)

3.3 While driving a vehicle, the fan-belt driving the alternator breaks.

3.3.1 What will indicate to the driver that this might have happened?

3.3.2 What effects could this have on the vehicle?

(2 × 1) (2)

3.4 List FOUR steps to follow when adjusting the v-belt tension. (4)

3.5 Name the part of the alternator that is identified as the output windings. (1)

3.6 Use the information in 3.6.1–3.6.2 below to identify the TWO types of advance systems found in a conventional ignition system.

3.6.1 It is controlled by the suction in the intake manifold.

3.6.2 It is controlled by the rotational speed of the distributor shaft.

(2 × 1) (2)

3.7 Name the tool that one should use to perform the following tasks:

3.7.1 To check the ignition timing

3.7.2 To set the contact breaker gap

3.7.3 To test the plug leads

3.7.4 To test the vacuum advance unit

(4 × 1) (4)

3.8 List THREE sensors of the EFI system that, when they become faulty, may cause the engine light to light up. (3)

3.9 Name ONE component of the secondary circuit of the conventional ignition system. (1)

[24]

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QUESTION 4

4.1 Use FIGURE 3 below to answer the questions 4.1.1–4.1.4 in the ANSWER BOOK.

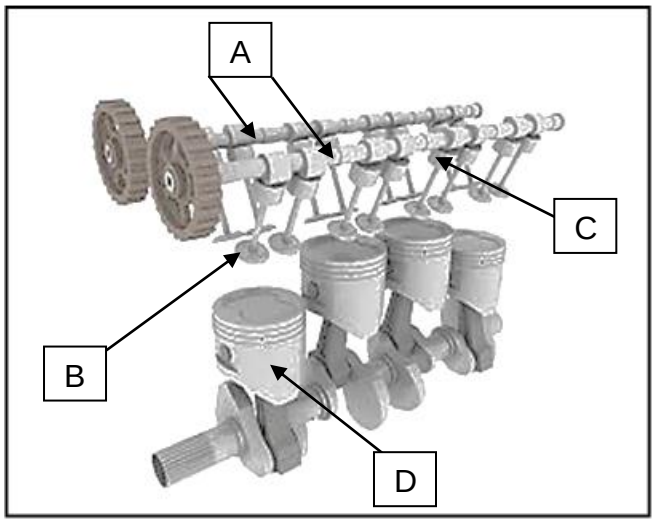


FIGURE 3

4.1.1 Name parts A, B and C in the correct order that they would be assembled. (4)

4.1.2 Name TWO measurements that can be done on part D to establish its condition. (2)

4.1.3 Identify TWO parts where the bearing journals must be checked for wear. (2)

4.1.4 Explain the procedure to have the crank bearing journals repaired if they are found worn after disassembly. (3)

4.2 Name the tools required to perform the following tasks to overhaul an engine:

4.2.1 Refit the pistons into the cylinder

4.2.2 Install the valves into the cylinder head

4.2.3 Tighten the bolts to the correct tension

(3 × 1) (3)

4.3 Incorrect storing of engines may be very costly.

Explain FIVE safety precautions to adhere to when storing an engine. (5)

4.4 Compile a list of FIVE inspections that one should perform on a vehicle after it had an engine fitted. (5)

[24]

TOTAL: 100

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